1. Calculate the penalty factor of a generating plant if increasing its output by 100 kW results in an increase of 10 kW in total system losses.
2. Discuss the impact of solar Energy Sources (RES) on system inertia. Why might this lead to the creation of "inertia markets"?
3.
    a. Why is voltage control required in power systems?
    b. Mention the different methods of voltage control employed in power system.
    c. With a neat connection diagram, explain one method of voltage control.
4. Find the optimal power allocation (neglecting losses) for two thermal units with the following cost functions: $C_1 = 25P_1 + 0.25P_1^2$ and $C_2 = 35P_2 + 0.25P_2^2$, to serve a total load of 200 MW.
5. Contrast primary, secondary, and tertiary frequency control in terms of their operating timescales and specific objectives.
6. Two interconnected areas 1 and 2 have the capacity of 250 and 600 MW, respectively. The incremental regulation and damping torque coefficient for each area on its own base are 0.3 and 0.07 p.u. respectively. Find the steady-state change in system frequency from a nominal frequency of 50 Hz and the change in steady-state tie-line power following an 850 MW change in the load of Area-1.
7. Determine the Area Control Error (ACE) for a system where the frequency has dropped by 0.05 Hz, the frequency bias factor *B* is 500 MW/Hz, and the outward tie-line flow has increased by 20 MW.
8. Compare the priority list method and dynamic programming for solving unit commitment. What are the trade-offs between computational speed and optimality?
9. Using the Newton-Raphson update logic, if the power mismatch ΔP is 10 MW and the sum of the gradients of the power output functions ($\sum {}^{1}/_{2d_i}$)is 100 MW per MWh, what should be the correction to the system lambda ($\Delta\lambda$)?
10. The recent Akosombo fire exposed a single point of failure in Ghana's transmission grid. From a power system operation and control perspective, discuss three specific technical measures — with supporting mathematical or operational justification — that GRIDCo can

implement to prevent a similar cascading failure. Your answer must reference concepts from:

i. N-1 security criterion
ii. Spinning reserve requirements
iii. Optimal power flow (OPF) and contingency-constrained dispatch

11. Calculate the new frequency of a system following a 50 MW load increase, given a set of generators with specific speed-droop characteristics (R) and a load damping constant (D).
12. Draw the block diagram of the LFC of a single-area system.
13. Compare the steady state and dynamic operations of an isolated system.
14. Draw the schematic diagram of a speed-governing system and explain its components on the dynamic response of an uncontrolled system with necessary equations. Hence, obtain the transfer function of a speed-governing system.
15. How do the governor characteristics of the prime mover affect the control of system frequency and system load?
16. Explain why it is necessary to maintain the frequency of the system constant.

17. On April 23rd, 2026, a major fire destroyed critical transformer infrastructure at the Akosombo 161 kV substation — a pivotal node in Ghana's national transmission grid operated by the Ghana Grid Company (GRIDCo). The substation serves as the primary evacuation point for power generated at the Akosombo Hydroelectric Dam (1,020 MW installed capacity) and the Kpong Hydroelectric Plant (160 MW), collectively supplying a significant share of Ghana's baseload power. The fire rendered multiple transformer bays inoperable, forcing emergency islanding, load shedding cascades across the southern interconnect, and emergency redispatch of thermal generation from the Aboadze Thermal Plant (TAPCO/TICO, ~550 MW combined) and CENIT Energy gas plants to compensate.

    Ghana's grid operates as part of the West African Power Pool (WAPP) interconnection, with cross-border power exchange agreements with Togo, Benin, Burkina Faso, and Côte d'Ivoire further complicating the post-fault system recovery.

System Data (Hypothetical):

If the pre-fault system has four major generating sources dispatched as follows:

Plant	Type	P_{Gi} (MW)	P_{min} (MW)	P_{max} (MW)
G_1 — Akosombo Hydro	Hydro	700	100	1020
G_2 — Kpong Hydro	Hydro	140	40	160
G_3 — Aboadze Thermal	Gas/Steam	200	50	550
G_4 — CENIT Gas	Gas	80	20	200

The incremental fuel cost characteristics of the thermal plants (G_3 and G_4) are:

$$\frac{dC_3}{dP_{G_3}} = 0.008P_{G_3} + 18.0 \quad (\text{GH₵/MWh})$$

$$\frac{dC_4}{dP_{G_4}} = 0.012P_{G_4} + 22.0 \quad (\text{GH₵/MWh})$$

The B-coefficient loss matrix for the post-fault reduced network is:

$$[B] = \begin{bmatrix} 0.0010 & -0.0002 \\ -0.0002 & 0.0015 \end{bmatrix} \text{ MW}^{-1}$$

(referenced to G_3 and G_4 only, following hydro evacuation loss from Akosombo substation outage).

The hydro scheduling constraint (water availability) limits combined hydro output post-fault to:

$$P_{G1} + P_{G2} \leq 500\ MW \text{ (emergency reserve dispatch limit)}$$

The total system demand at the time of the fault is 1,120 MW with a projected load growth rate of 2.5% per hour due to evening peak onset.

a. Prior to the fire, the system was on economic dispatch with all four plants online. Given that hydro plants have negligible incremental fuel costs (treated as must-run units fixed at their pre-fault outputs), verify that the thermal plants G_3 and G_4 were operating at their economically optimal outputs for a residual thermal demand of 280 MW. Determine the optimal system lambda (λ) and confirm each plant's dispatch.

b. Calculate the total thermal generation cost per hour at the optimal dispatch point determined in (a), given:

$$C_3 = 0.004P_{G_3}^2 + 18P_{G_3} + 500 \quad (\text{GH₵/hr})$$

$$C_4 = 0.006P_{G_4}^2 + 22P_{G_4} + 300 \quad (\text{GH₵/hr})$$